

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8523**

AV:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600063
Course Section:	1				

L_29_520_986_L1505_38388



Name of the candidate. Avani Vipat

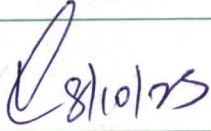
Candidate's UID number:

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Examination class room number:

4	0	1
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Invigilator's signature with date:



(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

PROGRAM: CSE-AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: DAY-Wednesday DATE: 08/10/2025

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.1a. $f(w) = \frac{1}{2} \|Xw - y\|^2$

$$X^t = X^{t-1} - \eta \nabla f(w)$$

We do partial derivative of the function and multiply it with the learning rate.

$$\frac{\partial f(w)}{\partial (X)} = 2 \times \frac{1}{2} \|W\|$$

$$= \|W\| = w$$

$$X^t = X^{t-1} - \eta w$$

$$\eta w = X^{t-1} - X^t$$

$\lambda_{\max}$ = largest eigen value of $X^T X$.

$$|A - \lambda I| = 0$$

$$A - \lambda I = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} a_1 - \lambda & a_2 \\ a_3 & a_4 - \lambda \end{bmatrix}$$

$\therefore$ minimum learning rate value should be $\frac{2}{\lambda_{\max}}$



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Q.16 Bias-Variance decomposition for expected squared error of a predictor.

$$W_f = \frac{1}{n} \sum (\hat{y}_i - y_i)^2 + b$$

where $\hat{y}$ = mean

y_i = predicted values

b = bias.

n = no. of samples.

When the model's capacity becomes too high, it tends to train on more and more samples which cause high learning of too many features, unnecessary noise is also been trained, this causes overfitting i.e. high variance, so we need to reduce the variance to overcome overfitting.

Since it also records the noise, hence causing problems in generalization.



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	Q.2a	The convolution^{computational} complexity of a standard 2D convolution is $[n \times n]$ where n is the size of matrix. $t_c = \sum_{n \times n} n \times n$ & of a depthwise separable convolution, for input size $H \times W \times C$ with K filters is $t_{cd} = \sum_n \sum_{i=0}^K H \times W \times C$ where $n = F \times F$. The depthwise separable convolutions provide efficiency in terms of parameters since the convolution is done in depth, the parameters are optimized hence it is separated after the convolution is optimized convolved through depth. Hence in modern architectures, where the architecture is too complex, it is efficient to use depthwise separable convolutions. When convolved in depth, all the parameters are optimized providing an efficient system.

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Q2b Autoencoder, encodes the input data detects the high contrast points, and ~~the~~ decodes further to give the predicted value.

In PCA, the important components are extracted from the image removing noise, gives necessary features / components

An under complete autoencoders only encodes on a smaller scale, not giving fully decoded output. Hence it is similar to the PCA or Principal component analysis where only required part is extracted, not efficient, though, it helps in dimensionality reduction.

An overcomplete autoencoder gives precise output by encoding each and every pixel in the input image and decodes it where ~~it~~ the output is very sharp, giving over accurate results. Hence we need to regularize it. Since overcomplete autoencoder, will ~~se~~ encode the output image such that the encoded image will only contain binary objects tending to learn identity function.

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Q3a

recursive gradient expression
for RNN $= \frac{\partial L}{\partial h_{t-k}}$

$$x^t = x^{t-k} * -\eta \nabla f(h)$$

where $\nabla f(h)$ is the partial derivative of
hidden state.

$$\therefore \nabla f(h) = \frac{\partial L}{\partial h_{t-k}} \leftarrow L \text{ is the Latent Space.}$$

~~The forget gate in LSTM only ~~rem~~
remembers the necessary words like
noun, adverb etc,~~

The forget gate in LSTM, ~~rem~~ forget
the parts from the input which are not
important, remembering only important
parts in the sentence / input like noun
adverb etc.

In this way it modifies recursion
since only some part remembered
else is lost, ~~as~~

it uses tanh function (LSTM).

Unlike in RNN, where after some point
information is lost

The input comes to the LSTM, goes into
tanh, remembers necessary details, part
to be allowed to forget is passed to
forget gate, which decides to forget
the unimportant.

the gradients in between hidden layers
are prevented from vanishing.
under suitable initialization

Sr. No. 88788523



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